WORKED SOLUTION

STEP II 2002, Question 11

Mechanics • Friction and Limiting Equilibrium

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q11

The key idea

The first part is a direct centre-of-mass integral; a substitution $t = x/l$ makes it a one-liner. For the ladder, "limiting at both ends" turns all four contact forces into multiples of just one unknown, so resolving twice and taking moments about A gives $\tan\theta$ at once. For the final part we need one more idea: the beam is safer the steeper it stands, so "slides for every $\theta < \pi/4$" says the limiting angle is at least $\pi/4$.

Solution

The centre of mass

A small piece of length δx at distance x from B has weight $\alpha W l^{-1}(x/l)^{\alpha-1}\,\delta x$. With $t = x/l$ (so $\mathrm{d}x = l\,\mathrm{d}t$), the total weight is

$$\int_0^l \frac{\alpha W}{l}\left(\frac{x}{l}\right)^{\alpha-1}\mathrm{d}x = \alpha W\int_0^1 t^{\alpha-1}\,\mathrm{d}t = W,$$

as it should be (the integral converges since $\alpha > 0$). The moment of the weight about B is

$$\int_0^l x\,\frac{\alpha W}{l}\left(\frac{x}{l}\right)^{\alpha-1}\mathrm{d}x = \alpha W l\int_0^1 t^{\alpha}\,\mathrm{d}t = \frac{\alpha W l}{\alpha+1}.$$

Dividing by W, the centre of mass G_0 is at distance $\alpha l/(\alpha+1)$ from B, as required. So it is at distance

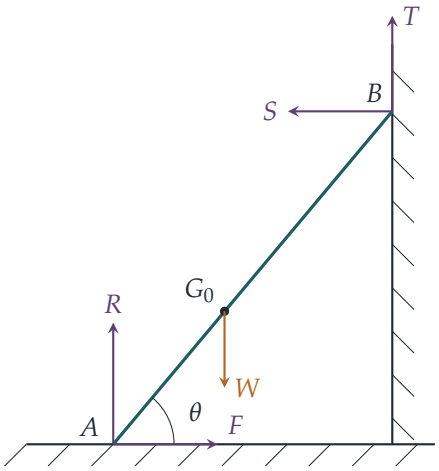
$$d = l - \frac{\alpha l}{\alpha+1} = \frac{l}{\alpha+1}$$

from A, which is what we will use for moments about A.

Limiting equilibrium at both ends

Model the beam as a rigid rod, with rough contacts at A and B. If it slips, A moves away from the wall and B moves down, so friction acts towards the wall at A and

upwards at B. Let the floor exert normal reaction R and friction F at A, and the wall exert normal reaction S and friction T at B. The weight acts at G_0, a distance $l/(\alpha+1)$ from A.



Resolving horizontally and vertically, and taking moments about A (the weight acts at horizontal distance $d\cos\theta$ from A; B is at height $l\sin\theta$ and horizontal distance $l\cos\theta$):

$$\begin{aligned} F &= S, \\ R + T &= W, \\ S\,l\sin\theta + T\,l\cos\theta &= W\,\frac{l}{\alpha+1}\cos\theta. \end{aligned}$$

Limiting at both ends means $F = \mu R$ and $T = \mu S$. Then $S = F = \mu R$ and $T = \mu S = \mu^2 R$, and the vertical equation gives

$$R(1+\mu^2) = W, \qquad \text{so} \qquad S = \frac{\mu W}{1+\mu^2}, \quad T = \frac{\mu^2 W}{1+\mu^2}.$$

Dividing the moment equation by $l\cos\theta$ and substituting,

$$\frac{\mu W}{1+\mu^2}\tan\theta + \frac{\mu^2 W}{1+\mu^2} = \frac{W}{1+\alpha} \quad\Longrightarrow\quad \mu\tan\theta = \frac{1+\mu^2}{1+\alpha} - \mu^2 = \frac{1-\alpha\mu^2}{1+\alpha},$$

that is,

$$\tan\theta = \frac{1-\alpha\mu^2}{(1+\alpha)\mu},$$

as required. Call this angle θ_0, and write $t_0 = \tan\theta_0$.

The greatest value of μ

Ask yourself Why does "slides for every $\theta < \pi/4$" translate into a condition on θ_0?

We show that equilibrium is possible exactly when $\tan\theta \geq t_0$. Write $t = \tan\theta$ and $k = 1/(1+\alpha)$, so the moment equation reads $St + T = kW$. The friction laws are $F \leq \mu R$ and $|T| \leq \mu S$.

Necessity. From $T \leq \mu S$ and $S = F \leq \mu R = \mu(W - T)$ we get $T \leq \mu^2(W - T)$, so $T \leq \mu^2 W/(1+\mu^2)$. Also $S \leq \mu(W - T)$, so

$$kW = St + T \leq \mu t(W - T) + T = \mu t W + (1 - \mu t)T.$$

If $\mu t < 1$, using the bound on T gives $kW \leq W(\mu t + \mu^2)/(1+\mu^2)$, which rearranges to $t \geq t_0$. If $\mu t \geq 1$ then $t \geq 1/\mu > t_0$ anyway, since $\mu t_0 = (1 - \alpha\mu^2)/(1+\alpha) < 1$. Either way, equilibrium forces $t \geq t_0$.

Sufficiency. If $t \geq t_0$ (and $t > 0$), take the wall friction limiting, $T = \mu S$, with $S = kW/(t+\mu)$ from the moment equation, and $F = S$, $R = W - T$. The only condition left is $S \leq \mu R$, i.e. $S(1+\mu^2) \leq \mu W$, i.e. $k(1+\mu^2) \leq \mu(t+\mu)$, which is exactly $t \geq t_0$. (Then $R > 0$ too.)

So the beam slides at angle θ precisely when $\tan\theta < t_0$. "It slides for every $\theta < \pi/4$" means $t_0 \geq 1$: if $t_0 \geq 1$, every $\theta < \pi/4$ has $\tan\theta < 1 \leq t_0$; if $t_0 < 1$, any angle with $\max(t_0, 0) < \tan\theta < 1$ is below $\pi/4$ yet in equilibrium.

With $\alpha = \frac{3}{2}$,

$$t_0 = \frac{1 - \frac{3}{2}\mu^2}{\frac{5}{2}\mu} = \frac{2 - 3\mu^2}{5\mu} \geq 1 \iff 3\mu^2 + 5\mu - 2 \leq 0 \iff (3\mu - 1)(\mu + 2) \leq 0,$$

multiplying by $5\mu > 0$. Since $\mu + 2 > 0$, this holds exactly when $\mu \leq \frac{1}{3}$. The greatest possible value is

$$\mu = \tfrac{1}{3}.$$

Checks

- With $\alpha = 1$ the beam is uniform and the formula becomes $\tan\theta = (1 - \mu^2)/(2\mu)$, the familiar result for a uniform ladder with equal coefficients at both ends.
- At $\mu = \frac{1}{3}$, $\alpha = \frac{3}{2}$: $t_0 = (2 - \frac{1}{3})/\frac{5}{3} = 1$ exactly, so the limiting angle is $\pi/4$. Numerically, $\mu = 0.332$ gives $t_0 \approx 1.004$ and $\mu = 0.334$ gives $t_0 \approx 0.996$.
- As $\alpha \to \infty$ the mass concentrates at B ($d \to 0$), and $\tan\theta$ becomes negative for any $\mu > 0$: the beam can stand at any angle, sensibly, since the weight then acts almost at the wall.

Common slips

- Using $\alpha l/(\alpha+1)$ as the distance from A in the moment equation; it is measured from B.
- Putting the wall friction downwards: when the beam slips, B slides down, so friction on the beam at B acts up.
- Reversing the final inequality: a *steeper* beam is safer, so "slides below $\pi/4$" needs $\tan\theta_0 \geq 1$, and the largest μ makes θ_0 exactly $\pi/4$.

Beyond STEP

The necessity-and-sufficiency argument above is really a small *linear programming* problem: the friction laws define a convex region of admissible forces, and equilibrium is possible when the moment line meets it. Statically indeterminate problems like this one, where the equations alone don't fix the forces, are a standard theme in first-year mechanics and engineering. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction, torque and rigid-body equilibrium with this kind of care.

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